

Expressions in Programming Languages

Concepts, Examples,

BAKD

DoCE

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What is an expression

- ✓ Operands and operators
- ✓ Operator precedence
- ✓ Associativity
- ✓ Short-circuit evaluation
- ✓ Side effects
- ✓ Type conversion (implicit & explicit)
- ✓ Best practice

What is an Expression?

- An expression is a combination of operands and operators
- Every expression evaluates to a value
- Expressions may have side effects
- Used in assignments, conditions, and function calls

Operands and Operators

- Operands: variables, constants, literals
- Operators: arithmetic, relational, logical, assignment
- Unary, binary, and ternary operators
- Operator behavior depends on language rules

Operator Precedence

- Determines order of evaluation
- Parentheses have highest precedence
- Multiplication before addition
- Assignment has low precedence

Precedence Example

- `int result = 10 + 5 * 2; // result = 20`
- `int value = (10 + 5) * 2; // value = 30`

Associativity

- Used when operators have same precedence
- Left-to-right or right-to-left evaluation
- Assignment operators are usually right associative

Associativity Example

- `int a, b, c;`
- `a = b = c = 10;` // evaluated right to left

Short-Circuit Evaluation

- Applies to logical AND (&&) and OR (||)
- Second operand may not be evaluated
- Improves performance and prevents errors

Short-Circuit Example

- `if(x != 0 && (y / x) > 2) {`
- `// Safe: division only if x != 0`
- `}`

Side Effects

- Occurs when expression modifies a variable
- Common with increment/decrement operators
- Order of evaluation matters

Side Effect Example (C)

- `int i = 5;`
- `int x = i++ + ++i; // Undefined behavior in C`

Type Conversion

- Implicit (type coercion) performed automatically
- Explicit casting done by programmer
- May cause precision loss

Type Conversion Example

- `int a = 5;`
- `double b = a;               // implicit conversion`
- `int c = (int) 3.7;   // explicit cast`

Best Practices

- Use parentheses for clarity
- Avoid expressions with multiple side effects
- Do not rely on evaluation order
- Write readable and maintainable expressions